

Sales Trend Analysis Using SQL Aggregations

Analyze monthly revenue and order volume from the online_sales dataset using SQL.

order_id	order_date	amount	product_id
1	2023-01-15	250.00	101
2	2023-01-20	150.00	102
3	2023-02-10	300.00	103
4	2023-02-25	200.00	104
5	2023-03-05	100.00	105
6	2023-03-20	400.00	106
7	2023-03-30	150.00	107

Query 1: Monthly Revenue and Order Volume

```
SELECT
  YEAR(order_date) AS order_year,
  MONTH(order_date) AS order_month,
  SUM(amount) AS total_revenue,
  COUNT(DISTINCT order_id) AS total_orders
FROM online_sales
GROUP BY YEAR(order_date), MONTH(order_date)
ORDER BY order_year, order_month;
```

Output:

order_year	order_month	total_revenue	total_orders
2023	1	400.00	2
2023	2	500.00	2
2023	3	650.00	3

Query 2: Top 3 Months by Revenue

```
SELECT
  YEAR(order_date) AS order_year,
  MONTH(order_date) AS order_month,
  SUM(amount) AS total_revenue
FROM online_sales
GROUP BY YEAR(order_date), MONTH(order_date)
ORDER BY total_revenue DESC
LIMIT 3;
```

Output:

order_year	order_month	total_revenue
2023	3	650.00
2023	2	500.00
2023	1	400.00